

Irin Rounds

Clinical Signal Cards for the Patient's Own Doctor

Two programs: Standing Cards and Step Watch

Impiricus Sponsor Challenge Specification

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Irin Rounds turns what happens at home into short, decision-ready cards for the patient's own doctor, delivered encrypted over Impiricus's verified HCP network, with pharma resources available only when the doctor asks. It runs two programs on one engine. Standing Cards watch an ordinary patient month after month. Step Watch watches the weeks after a therapy start, step by step, until the dose settles.

The problem. A CGM report shows what the numbers did and never why. The same rising overnight line can mean the basal dose is too low, the dose was taken late, dinner was under-covered, or the patient is rebounding from a low and the dose is actually too high. Those four need opposite prescriptions. On top of that, when a therapy changes the ground moves fast: the 2026 consensus tells clinicians to reassess insulin within 2 to 3 days of a GLP-1 start and at every dose step, using CGM data, while steps come every 4 weeks and visits do not. Between visits the doctor is blind, and the patient's only real options during a bad stretch are push through or stop.

How to Use This Document

This is a standalone handoff for a new Claude session or a teammate. It merges and supersedes both earlier briefs, `irin_rounds_impiricus_challenge.md` and the Step Watch brief. Appendix D records how the design got here.

Numbers marked "starting value" are tunable thresholds chosen during design, not researched facts. Numbers with a source are researched, and sources are in Appendix C. Market arithmetic mixes sources and years and is labeled as an estimate.

The team's repo rules in `CLAUDE.md` apply to everything here: ownership lanes, `contracts.py` as the treaty, `clock.py` for every timer, never commit raw CGM data, the existing safety invariants, main is human-only.

The main project documents (`Irin_Spec`, `CLAUDE.md`, `docs/plans/*.md`, the slash commands) were updated with this on 2026-09-25 (decision 26).

Freeze rule. Rounds work starts only after the core Irin demo runs end to end: The Save, both alarm tiers, replay, acknowledge.

Scope warning. The team chose to build both programs at the hackathon. That is the ambitious path. The build order and cut order exist so the project degrades gracefully instead of

collapsing; read them before writing code, and hold the line on the cut order when time runs short.

Terminology

The Irin_Spec terminology table still applies. These are the additions.

Term	Meaning
Clinical Signal Card	The published card format Rounds sends to a doctor. One schema, two programs, versioned as v0.
Standing Cards	The always-on program: Basal Check, Hypo Response, Follow-up. Fires on patterns, at most once per type per 14 days.
Step Watch	The therapy-start program: a scheduled watch that runs from a new prescription to a settled maintenance dose.
Clean night	A night with no late meal, no late correction, no exercise, basal on time, no treated low, adequate coverage, and the patient home. The only nights a basal conclusion can be drawn from.
Titration	Adjusting a drug's dose step by step until reaching the right amount. GLP-1s titrate UP every ~4 weeks; insulin usually titrates DOWN at the same time.
Maintenance dose	The dose where titration stops.
GLP-1 / GIP-GLP-1 RA	Weekly injectables (semaglutide, tirzepatide) that lower glucose and appetite. Main side effects are gastrointestinal.
Off-label	Used for a purpose not in the FDA label. Doctors may prescribe off-label; pharma may not promote it.
Nadir / low point	The lowest glucose in a period.
TIR / TBR	Time in range (70 to 180) and time below range (under 70). Consensus target for TBR is under 4%.
Level 2 low	Glucose under 54 mg/dL, clinically significant regardless of symptoms.
Nocturnal burden	Minutes under 70 in the night window, plus area under 70 in mg/dL-minutes.
Near-miss	A predicted-low warning that was acknowledged and the actual low never happened, usually because the patient

Term	Meaning
Unfelt low	treated it. Invisible to every CGM report. A low the patient did not notice, measured by asking the next morning about a specific event.
IAH	Impaired awareness of hypoglycemia. About 17.5% of adults with T1D, with roughly 4x the rate of severe lows.
HCP / PCP	Health care professional / primary care physician.
NRx / persistence	New prescriptions / the share of patients still on a drug over time.
Spark / Ascend / Wallet	Impiricus products. Spark triggers engagement from real-world events such as a first prescription; Ascend is the two-way AI inbox; Wallet sends copay cards, hub forms, and education to a patient's phone.
PDMA / BAA	The federal law governing drug samples / a HIPAA business associate agreement.

The Sponsor Challenge

Verbatim from Impiricus:

Challenge: Invent the Next Way We Engage HCPs. Build an HCP engagement tool Impiricus doesn't offer today or one built on top of what Impiricus already does. This can include net-new channels or new value layered on the existing platform and data. SMS and any existing features Impiricus currently offers or ships today are off-limits.

Your Mission: build a new HCP engagement tool; create a net-new channel or add new value on top of Impiricus's existing platform and data; do not use SMS or recreate any existing Impiricus features.

Judged on: Impact on the HCP. Originality. Technical execution. Commercial fit.

How Impiricus supports endocrinologists: by eliminating administrative friction around fast-evolving chronic care therapies such as GLP-1 medications, advanced insulins, and CGM devices, via (1) the Impiricus Wallet for copay cards, hub enrollment forms, and patient education; (2) on-demand PDMA-compliant sample fulfillment through QPharma inside Ascend; (3) immediate clinical decision tools including dosing calculators, drug safety information, and live routing to pharma support teams or field reps.

Two lines to anchor the pitch. Today every HCP engagement starts from pharma: a campaign, a claims trigger, a rep's calendar. Rounds is the first one that starts from the doctor's own patient. And the challenge text names GLP-1 medications first, which is exactly where Step

Watch aims. Impact on the HCP is judged first, so open with the doctor's problem and bring in pharma at the end.

What It Does

The Platform

- **One card format.** A Clinical Signal Card carries a headline, the metrics behind it, the nights it drew from, a narrative written from those numbers, and the actions the doctor can take. Published as JSON Schema so it reads as a platform, not a demo.
- **One encrypted channel.** Cards are encrypted on the device to the paired doctor's key. Impiricus relays ciphertext and a recipient ID and can read none of it. The doctor's browser decrypts inside Ascend.
- **Two programs on one engine.** Standing Cards for ordinary months, Step Watch for the weeks after a therapy start. Same night records, same metrics, same delivery, same inbox.
- **Numbers from code, words from the model.** Code computes every number; an LLM only phrases them; a validator rejects any number that is not in the card's metrics; a deterministic template is the fallback.
- **Nothing is ever prescribed.** No card contains a dose recommendation. The doctor types every number, and the patient confirms it out loud before anything changes.

Program A: Standing Cards

The always-on program for a patient who is not mid-titration.

- **Basal Check (detect).** Compares clean nights only. "Overnight rise on clean nights: +42 mg/dL median, 6 of 8 nights. 4 nights excluded: 2 late snacks, 1 rebound after a treated low, 1 late basal. Basal on time 13 of 14 days." Also fires in the "possibly too high" direction when near-misses accumulate.
- **Hypo Response (safety).** Median time to acknowledge nighttime alarms, warnings that escalated, re-arms, unfelt lows, counted only when the room confirmed presence. Paired with patient-reported glucagon status. "2 of 3 warnings escalated before acknowledgment, median response 7 minutes, one low not remembered, no glucagon on file."
- **Follow-up (verify).** Starts when the patient confirms a dose change. Compares clean nights before and after at day 7 and day 14. "Overnight rise went from +42 to +8." Says "not enough data yet" when clean nights are short.

Program B: Step Watch

The scheduled program for the weeks after a GLP-1, tirzepatide, or weekly basal insulin start.

- **One decision per dose step.** Three days before each scheduled step-up, a gate card: proceed or hold. Seven days after each step, a check card reporting what that step did.

- **Watches both directions.** Lows when insulin is not reduced fast enough, and highs or ketosis risk when insulin is cut too far or the patient stops eating.
- **Tolerance as a first-class signal.** A one-tap daily question, because nausea is why people quit, and a hold is the option that keeps them on therapy.
- **Ends.** Four green weeks at the maintenance dose and the watch graduates with a summary.

How the Two Programs Interact

They are deliberately not independent, because two programs messaging one doctor about one patient is how a good tool becomes noise.

- Starting a Step Watch **suspends Basal Check** for that patient. During titration a basal conclusion drawn from “clean nights” would be measuring the drug, not the dose.
- **Hypo Response stays live** during a watch but is merged into the watch’s red safety card rather than sent separately, so a patient never generates two safety messages for the same night.
- **Follow-up is the shared mechanism.** A dose change confirmed inside a watch starts the same before-and-after comparison that a standalone dose change does; the watch simply owns the reporting until graduation.
- At graduation, Standing Cards resume, and the first Basal Check after a watch uses the post-graduation nights as its baseline.

Safety Posture

Rounds is a companion to clinical judgment, not a medical device, and it never prescribes.

No card contains a dose recommendation. Irin never changes a prescribed schedule by itself; silence means the plan stands. Doctor messages never apply without patient echo-and-confirm. No conclusions from stale nights or from windows under 70% coverage. Patient data leaves the device only encrypted to the paired doctor. Every existing Irin invariant continues to hold, including that alarm logic never reads presence.

A Year With Irin Rounds (The Patient’s Perspective)

What the product is like to live with, no technical terms. This is also the story the pitch tells.

Pairing, once. At a visit, your endocrinologist taps “invite to Rounds.” Your device shows a code, she scans it with her phone, both screens show the same four digits, and you confirm. Nothing else changes. Your device still sits on the nightstand showing your number and glowing calm.

Ordinary nights. You log your basal when you take it, carbs when you eat, and the device does the rest. Some nights you eat late. Some nights you correct at bedtime. Some nights you go for a run at seven. Irin notices all of it, and quietly sorts your nights into the ones that can say something about your basal dose and the ones that cannot.

2:47 AM, a save. The forecast crosses your low line while you are asleep. Amber light, a soft tone, and you wake before the low instead of during it. Juice, acknowledge, back to sleep. In the morning your app asks one question: “At 2:47 AM you were 58 for about 25 minutes. Do you remember that?” You tap “I don’t remember it.”

Two weeks in. You never see this part. Your doctor gets a card: on your eight clean nights, you rose a median of 42 mg/dL overnight, six of eight in the same direction, and four nights were set aside with the reasons listed. She increases your basal by two units. Your phone and your device both ask you to confirm, out loud or with a tap, and you do.

Day 7 and day 14. She gets a second card saying the rise went from 42 to 8, and she does nothing, which is the correct action.

Months later, a new drug. She adds a weekly injection to your insulin and explains your insulin needs will drop, maybe by a third, and that she will not see you again for three months. She taps “start a Step Watch.” Your phone shows the plan: 2.5 mg weekly, first increase in four weeks.

The first weeks. Every morning your app asks how your stomach is: fine, rough, or can’t eat. One tap. Once a week you say “I took my shot” and confirm what it repeats back. On day 3 your doctor gets a card saying nothing has moved yet, and does nothing.

The step up. At four weeks you go to 5 mg. You are eating less because food stops being interesting, and the insulin dose that was right last month is now slightly too much. You are low at 3 AM twice that week, and both mornings you answer that you do not remember it.

Day 7 of the step. Your doctor sends back two things: a smaller basal dose, and the next increase pushed out four weeks. Both surfaces ask you to confirm and you do. The timeline on your screen slides: 7.5 mg moves from October to November.

What did not happen. You did not quit. You did not have a severe low at 3 AM. You did not spend three months deciding on your own that the drug was making you sick.

Graduation. You reach the dose that works and hold it for four quiet weeks. The watch ends with a summary your doctor reads before your next visit. If the device was a clinic loaner, you hand it back, and the standing cards carry on from your phone.

The Same Year (The Doctor’s Perspective)

Her inbox in Ascend is mostly quiet. Four patients on Rounds, one digest line reading “no action needed.”

The Basal Check. A card with a strip of fourteen nights, colored by reason: eight clean, two late snacks, one rebound, one late basal, two with poor sensor coverage. A chart of the rise on the clean nights only. The number: median +42, six of eight in the same direction. She does not have to decide whether a dip was treated or a spike was dinner, because the card already excluded those nights and told her why. Thirty seconds, two units, next patient. Two weeks later a Follow-up card tells her it worked.

The near-misses. On a different patient, the card says something no CGM report can: three predicted lows were caught and treated before they happened. That patient's time-below-range looks perfect precisely because the device keeps working. Her basal may be too high, and she would never have seen it in Clarity.

The Step Watch. Months later she starts the same patient on a GLP-1 and taps the offer that appears in her inbox. Steps arrive as a timeline, green until step 2, where an amber card shows the overnight low point down 22 mg/dL, time below range at 4.0%, two near-misses, one low the patient does not remember, and rough stomach three days of five. Every number carries a label saying whether it was measured, reported, or inferred. She types a new basal number, taps hold for four weeks, and moves on. Later that morning: "patient confirmed 07:14."

When she needs something from a manufacturer. On the Hypo Response card there is no glucagon on file. She taps access and resources, picks the category, then the brand, and Ascend does what Ascend already does. A banner says no patient data is shared with any manufacturer, and it is true: the relay only ever held ciphertext.

How It All Works

Irin Brain and Irin Bedside

Irin Brain is the software: night classification, per-night metrics, the glucose-only forecaster, both card programs, the narrative writer and its validator, and delivery. The patient uses it through the phone app.

Brain alone produces: reason codes (logged where the patient logs, inferred otherwise), overnight rise on clean nights, low-point shift, TBR, nocturnal burden, level 2 lows, ketone-risk episodes, TIR, mean glucose change, coverage, tolerance, adherence, unfelt-low rate, suspected unfelt lows inferred from the recovery slope, and predicted-low warnings delivered as phone notifications with their interaction timestamps.

Brain alone cannot: reliably wake someone, since a phone buzz is the alarm people sleep through; confirm a response time, since a slow acknowledgment may mean an empty room or a phone in the kitchen; or explain a night with logged meal and basal-timing context when the patient logs nothing.

Irin Bedside adds room-scale LED and sound alarms, step-week vigilance, confirmed near-misses and response times counted only when presence says home, hands-free voice logging, logged context, and the confirm screens.

Of the signals a card uses, the large majority come from the CGM or the phone. Two need the room. Say it that way round.

Where Brain runs without a Pi. Three options. Inside the phone app, which keeps encryption at the source but needs the model ported to JavaScript or ONNX and relies on unreliable background scheduling. A patient-owned cloud instance, the way the community already runs Nightscout on their own Railway account, which is reliable and patient-controlled and moves the encryption boundary to their own server. Or hosted by Irin, which is simplest and has the weakest privacy story. The pitch answer is the second. At the hackathon, Brain

runs on the Pi with a `brain_only` flag that ignores alarm events, presence, and logged context, so both tiers can be shown on the same data. Added 2026-09-25: no Brain-only backend is deployed for the hackathon; Brain-only is demonstrated from the Pi's adapter with `IRIN_BRAIN_ONLY=1` (or the demo-panel toggle). Later it could run as a container in `deploy/docker-compose.yml` on the Vultr box beside the relay and Irin Cloud; either way the My Irin dashboards the patient sees come from Irin Cloud, a separate patient-owned service the relay never touches.

Evidence Tiers

A card never shows a blank row labeled “requires Irin Bedside.” That reads as a paywall and undersells what glucose and a phone prove. Every metric carries a confidence label instead, which matches Irin’s existing honesty rule about stale data.

Label	Meaning	Example
Measured	Recorded directly by the device	Alarm fired 2:47, acknowledged 2:51, presence home
Reported	The patient answered a question about it	“I don’t remember that low”
Inferred	Derived from the shape of the glucose curve	Recovery slope 0.53 mg/dL/min with nothing logged, so likely untreated

The clinical frame: **Irin Brain is the screen, Irin Bedside is the confirmatory test.** Brain runs on every patient a clinic enrolls, at no cost, and finds the nights that do not add up. Bedside answers the one question the curve cannot: did she wake, when, and was she there. Like an ECG leading to a Holter monitor, and temporary in the same way during a titration, because the device can be a clinic loaner returned at graduation.

Two lines to say out loud: “The app tells you whether she knew. The device tells you whether she woke, when, and that she was there.” And: “Nothing on the card is blank. Every number carries how sure we are.”

The Card Pipeline

One path, whichever program is asking.

1. **Night ledger.** At night-window end the engine writes a NightRecord: coverage, reason codes, rise, low point, nocturnal burden, near-misses, level 2 count, alarm event IDs.
2. **Low events and recall.** Each nocturnal low becomes a LowEvent with its nadir, duration, recovery slope, and inferred-unfelt flag. In the morning the app asks about it.
3. **Programs evaluate.** Standing Cards check their pattern rules over the last 14 nights. Step Watch checks whichever window just closed. Both are pure functions taking records and returning status, flags, metrics, and confidence labels.
4. **Noise budget.** Rate limits and program interaction rules decide what actually sends.
5. **Narrative.** Numbers from code, words from the model, validator, or template.

6. **Delivery.** Encrypt to the doctor's key, POST ciphertext to the relay, email alert carrying no patient data.
7. **Reply.** The doctor's action returns as an authenticated message; the patient echo-confirms; the change applies; a receipt goes back.

The Step Watch Lifecycle

1. **Trigger.** The doctor writes a new GLP-1 prescription. Spark already detects first-time prescriptions from claims, so Ascend shows "New GLP-1 start. Offer Step Watch?" A doctor can also start one manually.
2. **Plan setup, about 30 seconds.** Drug class and schedule pre-filled from the label and editable, start date, whether the patient is on insulin, and options: ketone prompts (default on for T1D), step-week vigilance (default on for insulin users), slower escalation every 8 weeks.
3. **Invite and pairing.** Already paired patients skip this. Otherwise the device shows a QR code, the doctor scans, both sides show a 4-digit code, and the patient confirms with the PIN.
4. **Plan delivery.** The plan arrives as an encrypted, authenticated message, and the patient confirms it before the watch is active.
5. **Baseline.** The 14 nights before the start, already in the database. Fewer than 5 usable nights is reported as "baseline thin."
6. **Early check on day 3.** The consensus's "within 2 to 3 days," evaluating nights 1 to 3, labeled as limited data.
7. **Step check on day 7 after each step,** evaluating days 3 to 7.
8. **Step gate 3 days before each planned step-up.** Proceed or hold.
9. **Silence means the plan stands.** Irin never blocks a prescription.
10. **Doctor acts.** A hold shifts every later date; an insulin change updates settings and logs a therapy change; both need patient confirmation.
11. **Safety bypass.** A level 2 low, a re-armed alarm, or a ketone-risk episode on a "can't eat" day sends a red card immediately.
12. **Graduation.** Four consecutive green weeks at the maintenance dose. The final card compares baseline to now and doubles as visit prep. Standing Cards resume.

Watch length. Tirzepatide runs 2.5, 5, 7.5, 10, 12.5, 15 mg, so 5 step-ups; at 4 weeks each, 15 mg arrives at $5 \times 4 =$ week 20, and graduation is week 24. Ozempic has 3 step-ups, so 2 mg at week 12 and graduation at week 16. Wegovy has 4 step-ups and graduates around week 20.

Who It Serves

Segment	Size (estimate)	Programs	Pharma resources rail
Type 1 on injections	~2.1M, minus pump users	Standing Cards always; Step Watch if a GLP-1 is added	Glucagon and access categories on; GLP-1 rail off, since glucose use is off-label
Type 2 on insulin	~6.3M	Both, and the strongest commercial	On, and on-label

Segment	Size (estimate)	Programs	Pharma resources rail
Type 2 not on insulin	~20.7M	segment Step Watch in its tolerance-and-response form, Brain only	On

Arithmetic, mixing a 2026 prevalence figure with a 2022 insulin figure, so an estimate: diagnosed diabetes about 29.1M; type 1 about 2.1M; insulin users about 8.4M; type 2 on insulin $8.4 - 2.1 =$ about 6.3M, roughly 3 times the entire type 1 population; type 2 not on insulin $29.1 - 2.1 - 6.3 =$ about 20.7M.

The HCPs are endocrinologists and, for most type 2 patients, primary care physicians, who arguably gain more per card. Lead the commercial story with type 2 on insulin, the one segment where the product, the device, the on-label rules, and pharma's money all line up.

Growth loop. A clinic turns Rounds on for every CGM patient it follows, free and with no hardware. Cards flag the patients whose nights are ambiguous or risky. The doctor recommends Irin Bedside to those patients, and their cards become full-context. That answers “who buys the device?” without requiring anyone to.

Impiricus's Role

In Their Own Words

Impiricus's CEO and co-founder, Dr. Osama Hashmi, a practicing dermatologist, makes the same point in two separate 2026 announcements:

“Physicians should be able to connect patients to critical treatment resources the moment they are needed, without navigating fragmented systems or administrative barriers.” (Medvantx integration, September 2026)

“Physicians should be able to access critical resources for patients the moment we need them, without navigating fragmented systems or delayed follow-up.” (QPharma integration, August 2026)

And on the Ascend launch: “Ascend gives doctors choice and relevance in how they engage, while giving pharma a responsible way to share information.” The stated mission is ethically connecting HCPs to pharma resources and accelerating patient access, with every interaction staying clinically meaningful.

The turn. That quote is about the moment. Everything Impiricus ships today assumes the doctor is already in the conversation, because the moment is a visit, a rep interaction, or a claims trigger that arrives weeks later. Nothing on the platform can tell a physician when the moment has arrived for a patient at home. Rounds supplies the moment. A fortnight of clean nights rising 42 mg/dL is a moment. Night 4 after a dose increase, when she was low for 25 minutes and did not know it, is a moment. Ascend already handles everything that comes after it.

Two more quotes for the pocket. Dr. Iqra Aftab, a rheumatologist, on the QPharma launch: “the best technology is the kind I hardly notice,” which is the argument for the digest and the rate limits. Dr. Kashif Firozvi, an oncologist, on Ascend: “This technology doesn’t just enhance communication, it builds trust,” which pairs with the “what Impiricus sees” panel.

Built on the Existing Platform

Impiricus asset	Role in Rounds
Spark, the first-prescription trigger	Offers Step Watch at the moment of a new start
Verified HCP network and directory	Delivery channel and the root of trust for doctor keys
Ascend inbox	Where cards live and actions happen, as a new card type
Wallet	Sends GI education, copay cards, titration guides, glucagon education, and the pairing invite to the patient
QPharma	Samples, including the next pen strength during titration, where the brand offers them
Medvantx	Bridge supply when coverage lapses mid-titration
Hub enrollment	Prior authorization when a higher strength needs it
Field reps and MSLs	Doctor-requested clinical questions

Framing slide: sources on the left (Irin, any CGM feed, future smart pens and pumps), the published Clinical Signal Card in the middle, the Impiricus relay, then the Ascend inbox and existing fulfillment. “Impiricus already owns the hardest part, the verified network. Rounds is an open inbound channel on it. Irin is the first device, not the only one.”

Business Flow, Patient to Pharmacy

1. The patient wears a CGM and lives their life. Impiricus not involved.
2. Every night Irin files a record locally. Impiricus not involved.
3. Pairing once, by doctor invite or device QR, with consent recorded and revocable.
4. A pattern holds, or a step window closes, and a card is built and encrypted.
5. **Impiricus:** the relay routes ciphertext to the verified doctor’s Ascend inbox, with an email alert carrying no patient data.
6. The doctor reviews; the browser decrypts; the review may support CGM interpretation billing.
7. The doctor decides. Clinical changes go back encrypted and authenticated, and the patient confirms. Prescriptions happen in the EHR as usual. **Impiricus:** resources are pulled by category, then brand.
8. **Impiricus:** fulfillment runs through existing partners. Pharma sees “HCP requested resource” and “copay card redeemed,” never a glucose value.
9. The loop closes with Follow-up cards, the next step, or graduation.

How Impiricus Earns From It

- **Pharma sponsorship on two different moments.** Standing Cards create prescribing moments that barely exist today, above all glucagon, where only 8.7% of short-acting insulin users filled it even once across a decade. Step Watch creates persistence moments during titration, where drop-off is the GLP-1 makers' largest problem. Impiricus already sells claims-measured lift; the new claim is a persistence lift measured the same way, comparing prescribers who use Rounds against matched prescribers who do not. Pharma receives no patient data of any kind.
- **Resource moments at real decision points.** A hold, a rough week, or a card showing no glucagon on file is exactly when a copay card, education, a sample, or a bridge supply changes the outcome.
- **Recurring non-promotional engagement.** A tirzepatide watch produces $5 \text{ step-ups} \times 2 \text{ cards} = 10$ cards, plus the early check, the baseline note, and graduation, so $5 \times 2 + 3 = 13$ clinically driven touchpoints across 24 weeks. An Ozempic watch gives $3 \times 2 + 3 = 9$ across 16 weeks. Standing Cards then continue at up to one card per type per 14 days indefinitely. That is clinical action between rep visits, which is Impiricus's own stated goal.
- **A second customer type: connected device makers.** CGM makers (Dexcom including Stelo, Abbott), pump and AID makers, smart pens and caps, and data platforms all need doctors to act on their data. Their portals are pull, where a doctor logs in and digs. Rounds pushes decisions into a channel doctors already use, including the roughly 30% of digitally dark HCPs those portals never reach. "Portals hold the data, Rounds delivers the decisions."
- **HCP intent signals.** "This prescriber actively manages titration" and "this prescriber requested glucagon access" are HCP-side engagement data, the class Impiricus already uses.

Resource categories: glucagon access, GI side-effect education, copay and savings, samples including the next pen strength, bridge supply, prior authorization and hub enrollment, ask an MSL. Category first, doctor picks the brand, always pulled, never pushed, labeled as manufacturer resources, hidden for off-label use.

Why Impiricus would adopt it. Not as "a bedside device with an Impiricus button," because hardware does not scale to a million HCPs and patient data is a liability. Plausibly as "an inbound clinical-signal card type in Ascend, triggered by their own Spark, aimed at their clients' largest drug class, paid for by pharma and device makers, holding no readable patient data, with hardware optional."

Challenge Rules Check

Rule	How Rounds meets it	Watch out
A new HCP engagement tool	Decision-ready clinical cards for the prescriber	Call it an engagement tool, not a clinic dashboard
Net-new channel or new value on the platform	Both: a new device-to-doctor encrypted channel, layered on Spark, Ascend, Wallet, and the fulfillment partners	None

Rule	How Rounds meets it	Watch out
No SMS	Cards live in the Ascend inbox; the email alert carries no patient data	Ascend partly runs over SMS, so never say cards go out by text
Do not recreate features	Hands off to Wallet, QPharma, Medvantx, hubs, and MSLS	No dose numbers on cards, which would look like the dosing calculators

Signals and Card Rules

All thresholds are starting values, to be tuned against Chris's history.

Night Classification

Every night gets one or more reason codes. Night window defaults to 22:00 to 07:00.

Reason code	Rule
late_meal	Carbs logged within 3 h before night start, or during the night
late_correction	Bolus within 3 h before night start, so IOB above zero at night start
basal_late / basal_missed	Basal logged more than 60 min after the usual time, or not logged
exercise	Exercise logged after 17:00
treated_low	Any low alarm followed by a rise above about 60 mg/dL within 2 h
stale	Coverage under 85% of expected readings
away	Presence toggle set to Away
clean	None of the above

Coverage arithmetic: $9 \text{ h} \times 12 \text{ readings/h} = 108$ expected, and $0.85 \times 108 = 91.8$, so at least 92 readings.

Codes are labeled by source. On a patient who logs, they are **logged**. On a Brain-only patient, late_meal is **inferred** (a rise faster than about 2 mg/dL/min in the first 2 h of the night), treated_low is **inferred** (a reading under 70 followed by a rise above 60 within 2 h), stale is fully available, and basal_late, exercise, and away are **unavailable** and shown as “context unavailable” rather than assumed clean.

Shared Signals

Signal	Definition
Overnight rise	Glucose at 07:00 minus glucose at night start plus 2 h, each a median over a 15-minute window to damp noise
Dawn rise	03:00 to 07:00, same method

Signal	Definition
Nightly low point	Minimum of the 15-minute rolling median in the night window
Low-point shift	Median nightly low point in a window minus the baseline median
TBR	Percent of readings under 70, full day and overnight separately
Nocturnal burden	Minutes under 70 in the night window, plus area under 70 in mg/dL-minutes
Level 2 lows	Any reading under 54, confirmed by 2 consecutive readings
Near-misses	Predicted-low warnings acknowledged or recovered with no actual crossing within 60 min
Suspected unfelt low, inferred	A nocturnal run under 70 lasting at least 20 min whose recovery slope over the 30 min after the nadir is under 1.0 mg/dL/min, with no carbs logged within 30 min
Unfelt-low rate, reported	Events answered “didn’t feel it” or “don’t remember” divided by events answered
Response time, Bedside only	Alarm start to acknowledgment, counted only when presence says home
Ketone-risk episode	A run at or above 200 mg/dL lasting at least 120 min
Tolerance	Counts of fine, rough, can’t eat, and missing days
Adherence	Logged weekly injections against expected, plus a dose-mismatch flag
Coverage	Readings received divided by expected

Standing Card Rules

- **Basal Check.** In the last 14 nights: at least 5 clean nights, a median rise beyond ± 30 mg/dL, and at least 70% of clean nights in the same direction. The “possibly too high” direction fires when near-misses reach 3 or more in 14 days. Never from stale nights. Actions: adjust basal, schedule visit, ask patient, dismiss.
- **Hypo Response.** In 14 days: at least 2 escalated warnings, or at least 1 re-arm, or a median acknowledgment time above 5 minutes, or any unfelt low reported. Adds glucagon status. Actions: message patient, schedule visit, access and resources, dismiss.
- **Follow-up.** Starts when the patient confirms a dose change. Compares the 14 nights before against days 1 to 7 and 1 to 14 after, requiring at least 3 clean nights each side, otherwise “not enough data yet.”
- **Rate limit.** One card per type per patient per 14 days.

Step Watch Windows and Status Rules

Windows. Baseline is the 14 nights before the start with night coverage at or above 85%, needing 5 such nights. Early check covers nights 1 to 3. A step window is days 3 to 7 after each step, where GI effects and glucose shifts cluster. The gate window runs from the end of the last step window to 3 days before the next planned step. Graduation is 4 consecutive weeks at the maintenance dose.

Status, evaluated in order:

1. **Red, any time and any coverage:** a level 2 low, a re-armed low alarm, or a ketone-risk episode on a day logged “can’t eat.” Sent immediately, deduplicated per event, at most one red card per type per 12 hours.
2. **Insufficient:** window coverage under 70%. No conclusions drawn; red events still go out separately.
3. **Amber, lows:** low-point shift at or beyond -15 mg/dL, or near-misses at or above 2, or TBR above 4%.
4. **Amber, awareness:** at least one nocturnal low reported unfelt or not remembered, or at least two inferred suspected unfelt lows with no answers.
5. **Amber, tolerance:** rough or can’t eat on at least 3 of 5 days. Hold becomes the prominent action.
6. **Amber, highs:** at least 2 ketone-risk episodes, suggesting insulin cut too far or the patient not eating.
7. **Green:** none of the above, and it goes to the weekly digest.

Rate limit. One step check and one step gate per step, with red exempt.

Doctor actions on a watch card: proceed, hold for 2, 4, or 8 weeks, adjust insulin with typed units and a start date, message, schedule visit, access and resources, end watch.

Worked Examples

Basal Check fires. 8 clean nights of 14, 6 rising. $6 \div 8 = 75\%$, which is at or above 70%. Median rise +42, which is beyond +30. The card fires and lists the 6 excluded nights with their reasons.

Step Watch amber, the demo card.

- Coverage: 5 days $\times$ 288 readings/day = 1,440 expected. The floor is $0.70 \times 1,440 = 1,008$. Received 1,390, and $1,390 \div 1,440 = 96.5\%$, so it passes.
- Low-point shift: baseline median 98, window median 76. $76 - 98 = -22$, at or beyond -15 , so it fires.
- TBR: 58 readings under 70. $58 \div 1,440 = 4.03\%$, above 4%, so it fires.
- Near-misses: 2, which meets the threshold, so it fires.
- Awareness: 1 nocturnal low answered “don’t remember,” so it fires.
- Tolerance: rough on 3 of 5 days, which meets 3 of 5, so it fires.

Headline: “Step 2 (5 mg), days 3 to 7. Overnight low point down 22 mg/dL from baseline, 2 near-misses, time below range 4.0%. One low not remembered. Rough stomach 3 of 5 days.”

Unfelt-low rate. 4 nocturnal lows in the window, 3 answered, 2 of those “don’t remember.” $2 \div 3 = 67\%$. The row reads “4 nocturnal lows in 14 nights, 3 answered, 2 unfelt (67%), 1 no answer.”

The Morning Recall Question

Why it matters. Impaired awareness of hypoglycemia is assessed today entirely by self-report questionnaires, asked once in clinic about a vague multi-month impression. The two standard instruments disagree about the same patient in nearly 60% of cases, and a 2025 review is built around how hard this is to assess in the CGM era. Nobody ships an event-level measure. Irin can, because the CGM already knows exactly which lows to ask about.

1. Overnight, the engine records a nocturnal low event, under 70 and confirmed by 2 consecutive readings.
2. In the morning the app asks, one card per event, capped at 2: “At 2:47 AM you were 58 for about 25 minutes. Do you remember that?”
3. Answers: felt it and treated it, woke but felt nothing, don’t remember it, or was awake anyway.
4. If carbs were logged within 30 minutes of the nadir, it pre-fills “treated” and asks only about symptoms.
5. Unanswered by noon is recorded as “no answer,” never as “fine.” Ask once, because recall decays.

Limits, stated plainly. Self-report gives no timing and cannot be checked. A patient with a reason to keep driving has a reason to say they felt it, and someone who half-woke may not know. That is exactly what the device measures.

Pitch line. “The two standard questionnaires disagree about the same patient nearly 60% of the time. We don’t ask her how she’s been doing. We ask about 2:47 last night, and we already know what her glucose did.”

Noise Budget

The whole product dies if doctors mute it. The rules, in one place:

- One Standing Card per type per patient per 14 days.
- One step check and one step gate per step.
- Green never interrupts; it goes into a weekly digest.
- Red bypasses everything, but is deduplicated per event and capped at one per type per 12 hours.
- A Step Watch suspends Basal Check and absorbs Hypo Response.
- Nothing at all is sent from stale nights or insufficient windows.
- A patient’s cards are never split across two programs on the same day.

Privacy, Pairing, and Trust

- **Blind courier.** Cards are encrypted on the device to the paired doctor’s key. The relay stores only ciphertext. The doctor’s browser decrypts.

- Where the words come from (added 2026-09-25). The narrative call runs on the device (or the patient's own Irin Cloud instance) and sends the computed metrics to the model provider through narrative.py's chain (Backboard, or the Meta Model API direct for the tasks NARRATIVE_ROUTING routes to provider meta, then the direct Claude call), exactly as the direct Claude call did; the relay never sees them, memory lives under the patient's assistant and is deletable through the API, and production would want BAA-covered inference. Say so if asked.
- **Hackathon pairing.** The device shows a QR code, the doctor scans it, and both sides display a 4-digit code derived from both public keys and the token, which defeats a key swap even by the relay. The patient confirms with the PIN.
- **Production pairing.** The Impiricus-signed doctor directory is the trust anchor; invites come from Ascend including telehealth short codes; private keys are protected by passkeys.
- **Authenticated replies.** Plan, hold, dose, and note messages must come from the paired doctor's key, and the patient still confirms.
- **What Impiricus sees.** A live panel in the demo shows the relay's whole view: recipient IDs, sizes, timestamps, ciphertext prefixes.
- **HIPAA, stated correctly.** Under HHS cloud guidance a service that stores encrypted health data is still a business associate even when it cannot read it. Impiricus is already HIPAA certified, so a BAA is routine. Encryption here is about patient trust, not about avoiding HIPAA.
- **Consent** is recorded at pairing, revocable from either side, and the patient can preview every card the doctor receives.
- **Why patient data never goes to pharma.** 64% of consumers will share health data with their own physician and 7% with tech companies; in a 24,000-person study the largest group was 97% willing to share with doctors and 9% with pharma, the least trusted recipient in every group.

Research Behind It

The core clinical insight. A rising overnight curve has at least four causes needing different actions: basal too low (increase), basal taken 2 to 3 hours late (a timing problem, not a dose problem), a late snack or under-bolused dinner (fix the bolus), or a rebound after treating a low (basal may be too HIGH, and increasing makes it worse). CGM-only tools cannot separate these, because patients on injections almost never log. A fifth case is created by Irin itself: when it warns at 2:47 and the patient drinks juice, the low never happens and the CGM report looks clean. **The better Irin works, the more the report misleads the doctor.** Near-misses are invisible to every CGM tool.

The 2026 consensus report on adjunct GLP-1 and GLP-1/GIP therapy in type 1 diabetes, published online 5 June 2026:

- Proactive insulin reduction is essential. Expect 25 to 35% lower total daily insulin in adults, about 20% in youth.
- Bolus reductions come earlier and are larger than basal reductions.

- Assess insulin needs within 2 to 3 days of initiation and reassess at every clinical encounter, using stepwise reductions to avoid ketosis.
- CGM is strongly recommended, and CGM data should supersede A1c during dose escalations, which typically occur every 4 weeks.
- Real-world studies have not shown a significant hypoglycemia increase, because clinicians reduce insulin proactively.
- Ketone monitoring when glucose is at or above 200 for 2 or more hours.
- Start low, go slow; some people need 2 to 3 month intervals.

Other findings:

- EASD 2025: daily insulin fell about 25% at 6 months on tirzepatide in T1D with obesity, both basal and bolus, without more hypoglycemia; nearly 90% were still on it at 6 months in a specialist-supervised cohort.
- Adding a GLP-1 to basal insulin is on-label in type 2, and labels advise considering an insulin or sulfonylurea reduction at the start.
- More than half of patients stop GLP-1s within a year; GI side effects, cost, and access lead the reasons.
- Insulin icodec (Awiqli) was approved 27 March 2026 for adults with type 2 only, U.S. launch expected in the second half of 2026.
- Glucagon: across 2011 to 2021, of 2.8M U.S. adults with diabetes only 1.5% filled glucagon; 8.7% of short-acting insulin users filled it at least once in the decade, against ADA guidance that it be prescribed for everyone on insulin or at high risk.
- Impaired awareness affects about 17.5% of adults with T1D, who report severe hypoglycemia at roughly 4 times the rate of others; the two standard questionnaires disagree in nearly 60% of cases.
- Nocturnal alarms are frequently ignored or silenced, and compression artifacts (mean about 45 minutes, mean drop about 24 mg/dL) cause false alarms.
- Therapeutic inertia in insulin titration is well documented.
- Billing, always “may support” and never promised: CPT 95251 once per 30 days at about \$33 Medicare, so $365 \div 30 = 12.17$, meaning 12 reviews a year and $12 \times \$33 = \396 per patient per year; 200 CGM patients $\times \$396 = \$79,200$ a year for a clinic. For a 24-week watch specifically, $168 \div 30 = 5.6$, so 5 reviews and about \$165. The 2026 codes 99445 and 99470 may also apply depending on payer.
- Competitive landscape: clinic triage by CGM metrics exists (Tidepool TIDE, Glooko population health, Dexcom Clarity clinic reports), none with behavioral context, alarm response, near-misses, or an access layer. Basal titration apps exist (Glooko MIDS, Insulia, My Dose Coach, d-Nav), all suggesting doses to patients, none routing a decision to the HCP. Roche’s Accu-Chek SmartGuide already offers MDI users a 30-minute low prediction, so “we predict lows” is not novel on its own. Irin’s edge is room-scale waking, context, two alarm tiers, and Rounds.

Value and Judging

Who	Gets	Pays
Patient	Problems caught between	Free for Brain; loaner or

Who	Gets	Pays
	visits, safer titration, data only to their own doctor	purchase for Bedside
Doctor	Decisions without detective work, 30-second reviews, possibly billable time	Free
Clinic	Revenue from reviews currently unpaid	Payers pay the clinic
Pharma	Prescribing and persistence moments at real clinical decisions	Pays Impiricus
Device makers	A route to verified doctors, including digitally dark ones	New Impiricus revenue line
Impiricus	Daily non-promotional engagement, real-time intent, a second customer type	Receives

Criterion	Estimate	Raises	Drags
Impact on HCP	8/10	Same line, opposite prescriptions; guideline-anchored titration; billable time	"More messages," answered by the noise budget
Originality	8 to 9/10	Near-misses; event-level awareness metric; patient-originated channel; post-prescription engagement	A judge who knows Clarity comparisons or Tidepool
Technical execution	8 to 9/10 if it runs live, 5 to 6 if mocked	Real hardware, a real model, real crypto, measured false-alarm rate and detection lag	Two programs is a lot of surface; synthetic titration data
Commercial fit	8/10	Glucagon moment, GLP-1 persistence, Spark trigger, on-label type 2, device makers	Persistence lift is a hypothesis; patient-data skepticism

Overall a serious first-place contender if the demo runs cleanly and the story fits the slot. The two biggest risks are judges getting lost across two programs plus crypto, and the build not

finishing. Both are handled the same way: one narrative (detect, decide, verify, then the pharma moment), and the cut order.

Tech Stack

Unchanged from Irin_Spec, plus: PyNaCl on the Pi and tweetnacl-js in the browser for end-to-end encryption; one small FastAPI relay service on the team's Vultr VPS, run from `deploy/docker-compose.yml` behind Caddy at `api.irin-out-of-sleep-at-hackgt.tech` with its documents on MongoDB Atlas (the same compose file on Chris's laptop is the fallback); the static clinician inbox served at `doctor.irin-out-of-sleep-at-hackgt.tech` by the same Caddy. No new ML dependencies, because every metric here is arithmetic over readings. Card narratives come from `narrative.py`'s chain, Backboard → Meta Model API → the direct Claude call → deterministic template, with the validator after every link; `NARRATIVE_ROUTING` sends `clinician_card` and `morning_report` to a Sonnet-class Claude and `second_opinion` to a third provider (openai), which answers `{invented_number, advice}` on every card; Backboard keeps per-scope memory. The device speaks its confirm screens and echoes in an ElevenLabs voice after the tone; Python 3.14 everywhere.

Interface Contracts

Added to `backend/app/contracts.py` in one announced commit, day one of Rounds work, never bundled into a feature branch. Exact Pydantic in Appendix A.

Object	Fields
NightRecord	<code>night_date</code> , <code>window_start</code> , <code>window_end</code> , <code>coverage_pct</code> , <code>reason_codes</code> , <code>code_source</code> (logged / inferred / unknown), <code>rise_mgdl</code> , <code>dawn_rise_mgdl</code> , <code>low_point_mgdl</code> , <code>tbr_pct</code> , <code>minutes_below_70</code> , <code>near_miss_count</code> , <code>level2_count</code> , <code>alarm_event_ids</code> , <code>is_demo</code>
AlarmEvent	<code>event_id</code> , <code>tier</code> , <code>started_at</code> , <code>acknowledged_at</code> , <code>ack_source</code> (device / app), <code>escalated</code> , <code>rearm_count</code> , <code>crossed_actual</code> , <code>presence_during</code>
LowEvent	<code>low_event_id</code> , <code>night_date</code> , <code>started_at</code> , <code>nadir_mgdl</code> , <code>nadir_at</code> , <code>minutes_below_70</code> , <code>auc_below_70</code> , <code>recovery_slope</code> , <code>carbs_logged_within_30min</code> , <code>inferred_unfelt</code> , <code>alarm_event_id</code>
LowEventRecall	<code>low_event_id</code> , <code>asked_at</code> , <code>answered_at</code> , <code>answer</code> (felt_and_treated / woke_no_symptoms / dont_remember / was_awake)
SymptomCheck	<code>date</code> , <code>gi</code> (fine / rough / cant_eat), <code>is_demo</code>
TitrationPlan	<code>plan_id</code> , <code>drug_class</code> , <code>drug_label</code> , <code>steps</code> , <code>started_at</code> , <code>on_insulin</code> , <code>options</code> , <code>status</code> , <code>is_demo</code>

Object	Fields
SignalCard	card_id, schema_version, program (standing / step_watch), kind (basal_check / hypo_response / follow_up / early_check / step_check / step_gate / safety / graduation), status, flags, confidence map, source (irin_bedside / irin_brain), plan_id, step_index, period, headline, metrics, nights, excluded_counts, tolerance_days, narrative, allowed_actions, resource_categories, is_demo, generated_at
Pairing	device_id, doctor_id, doctor_display_name, doctor_pk, status, confirmed_at
DoctorMessage	message_id, plan_id, kind (plan_create / plan_update / proceed / hold_step / insulin_change / note / schedule_request / end_watch / dismiss), hold_weeks, insulin, new_units, start_date, plan, text, created_at, status
Additions to existing objects	Treatment.kind gains glp1_dose and therapy_change, plus a dose_label; Settings gains basal_units, glucagon_on_hand, glucagon_expiry
New WebSocket messages	card_sent, doctor_message_received, doctor_message_resolved, symptom_check_due, recall_due, pairing_state, plan_state. The state snapshot gains the active plan, pending doctor messages, today's check-in and recall status, and pairing state

Contracts drift, noted 2026-09-25: Pairing carries peer_kind (doctor | buddy) and is_demo, AlarmEvent carries is_demo, and the card kinds include baseline_note; the rows above are the day-one list.

One card model serves both programs. The program field decides which rules produced it and which sections the inbox renders. Publish it as JSON Schema, "Clinical Signal Card v0," in relay/README.

Functional Requirements

Card Engine

- Writes a NightRecord at night-window end and evaluates both programs at 07:05, on clock.py so replay accelerates them.
- Derives what should have run from plan dates, night dates, and stored card IDs rather than from jobs that fired, so catch-up is idempotent across reboots and replay seeks.

- Rules are pure functions returning status, flags, metrics, and confidence labels, so the same code runs over history for validation.
- Red rules run on every new reading and every alarm-event update.
- Enforces the noise budget in one place, including program interaction.
- Never draws a conclusion from stale nights or windows under 70% coverage.
- Resource categories are empty for off-label use.
- In brain_only mode, Bedside metrics change confidence label; they are never blanked.
- The narrative comes from narrative.py with a clinician prompt of 2 to 3 sentences, no recommendations and no dose numbers; a validator checks every number against the metrics; a deterministic template renders on failure or with no network. The chain has four links (decision 29): Backboard first (assistant per doctor-patient pairing, memory Auto; NARRATIVE_ROUTING sends clinician_card to a Sonnet-class Claude, and a different provider, openai, answers {invented_number, advice} as the second_opinion, disagreement shipping the template), then the Meta Model API direct (the tasks routed to provider meta, or the openrouter rows when Backboard is down), then the direct Claude API, then the template; the validator runs after every link.

Patient App

The app carries all input; the device carries none.

- Daily stomach check-in during a watch: fine, rough, can't eat. Missing is "no data," never "fine."
- Morning recall questions for recorded nocturnal lows, capped at 2, expiring at noon.
- Logging: basal taken, carbs and units, exercise, notes, weekly injection, by voice or touch, with insulin always echoed and confirmed.
- Alarm acknowledge, the primary response measurement.
- Doctor-message confirm, mirrored on the device.
- Step timeline, card preview showing exactly what the doctor sees, pairing status and revoke, glucagon status toggle, settings.

Bedside Display

- Glance and alarm only: number, trend, forecast, step timeline, night clock, morning summary, DEMO badge, full-screen alarm with LED frame and sound.
- A physical acknowledge, because a phone across the room at 2 AM is the failure case the device exists to prevent. The alarm event records which surface acknowledged.
- A passive chip reading "2 questions waiting in the app."
- The doctor-message confirm screen and the ketone prompt banner.
- A "Share with my doctor" QR screen reached from settings.
- No forms anywhere.

Clinician Inbox

- Inbox list ordered red, amber, then a digest line for greens, with program, source, and DEMO badges, and Spark offers at the top.

- Offer page with the Step Watch plan setup form, pre-filled from a small JSON of label schedules.
- Pairing page reading the URL fragment and showing the 4-digit code.
- Standing card detail: a strip of nights colored by reason code, with logged and inferred distinguished, the rise chart on clean nights only, excluded nights with reasons, metrics, narrative, actions.
- Watch card detail: step timeline, low-point chart against the baseline band, metrics with confidence labels, tolerance strip, awareness row, narrative, actions.
- Action modals: adjust basal, hold 2 / 4 / 8 weeks, adjust insulin, message, schedule visit, dismiss, end watch.
- Resources flow: category, then brand, then a mock Ascend handoff carrying the banner “No patient data shared with any manufacturer.”
- A “what Impiricus sees” panel.
- Optional: tap a night to replay it, reusing the replay engine.

Relay

- Stores ciphertext only, with per-source API keys and rate limits.
- The Pi makes outbound requests only, so no new inbound port at home.
- Endpoints in Appendix B, including a mock Spark trigger for the demo and a resources-request endpoint whose schema has no patient field at all.
- Added 2026-09-25: hosted on Vultr from `deploy/docker-compose.yml` behind Caddy at `api.irin-out-of-sleep-at-hackgt.tech`; documents on MongoDB Atlas (cards, messages, pairings, hub, the Night Buddy directory, DevicePairing); the device key is the X-Source-Key header; `relay/notify.py` sends the Night Buddy WhatsApp message (never a glucose value); the laptop fallback is the same compose file with a local mongo container serving everything on one origin, `http://<laptop-ip>` (chris.md, R6 fallback). Readings never enter the relay; the patient’s time series and dashboards live in the separate Irin Cloud service.

Machine Learning and Measured Evidence

George’s existing pipeline is unchanged. Rounds adds `ml/models/nights.py`, the single shared night-classification and metrics module imported by both the backend engine and the validation scripts. Same rule as `features.py`: never implement a metric twice, because a definition that drifts between validation and serving fails silently.

Three numbers make technical execution credible, all from Chris’s real 22 months rather than the synthetic scenario:

1. **Retrospective lead time for Basal Check.** For each real basal increase Chris lists, run the Basal Check rule daily over the prior history and record the first trigger date. Lag equals the change date minus the first trigger. The claim becomes “Rounds would have flagged this N weeks earlier,” retrospective, n of 1, observational.
2. **False-alarm rate.** Run the Standing Card rules and the Step Watch amber rules over stable-therapy stretches, excluding gaps and any window within 30 days of a therapy change, and report how often they fire. Scale: 22 months is about $22 \times 30.4 = 669$ days, so

roughly $669 \div 5 \approx 133$ non-overlapping 5-day windows plus about 47 fortnights before removals.

3. **Detection lag.** Days from a real therapy change until the low-point shift rule first flags a shift of 15 mg/dL or more in either direction.

All three go in `ml/models/metrics.md` with `n` and caveats. Any model-based number on a demo window must come from the held-out 8 weeks or a retrained copy excluding that window, noted in `metrics.md`. Raw CSVs and the therapy-change list stay in gitignored `ml/data/`, and scripts print summaries only.

Team, Repository, and Workflow

Ownership and Lanes

Owner	Work
Chris	Contracts additions; the night ledger and alarm-event recorder; both programs' rules; low events and recall; narrative and validator; crypto, pairing, relay and relay client; the doctor-message path; replay seek and idempotent catch-up; tests; new invariants in the docs
George	nights.py classification and metrics; date-shifted demo windows around real basal changes; the synthetic titration scenario; retrospective lead time, false-alarm rate, detection lag; threshold tuning; metrics.md
Justin	The clinician inbox in full, both card layouts; pairing screen; patient app check-in, recall, logging, card preview, confirm screens; display timeline, chip, QR screen; demo panel jump-to-step buttons
Slavik	Optional message-waiting LED state, daytime only and never during an alarm; confirm the radar DEMO profile behaves with a judge at the table during pairing

Repository Additions

- `backend/app/rounds/`: `ledger.py`, `alarm_events.py`, `nights_adapter.py`, `low_events.py`, `recall.py`, `cards.py`, `standing.py`, `step_watch.py`, `narrative.py`, `noise.py`, `crypto.py`, `pairing.py`, `relay_client.py`, `messages.py`, `vigilance.py`
- `relay/`: the FastAPI relay plus README with the published card schema
- `frontend/clinician/`: the Ascend-styled inbox
- `ml/models/nights.py`: the shared classification and metrics module
- `demo/scenarios/`: date-shifted real windows plus `titration_synthetic.csv` and its companion JSON

Build Order

Both programs ship, but not in parallel from hour one. Standing Cards come first because they share the ledger, the card model, the delivery path, and the inbox; Step Watch then reuses all of it and adds the plan and the windows.

1. Contracts, ledger, alarm-event recorder, low events.
2. Delivery: crypto, pairing, relay, relay client, inbox shell.
3. Standing Cards: Basal Check first, then Hypo Response, then Follow-up.
4. The doctor-message path and patient confirm, proven once on a basal change.
5. Step Watch: plan, windows, status rules, gate and check cards.
6. Recall questions and the unfelt-low rate, which both programs use.
7. Resources handoff, the “what Impiricus sees” panel, the narrative validator.
8. Polish: step-week vigilance, night replay, Brain versus Bedside side by side.

Cut Order

First cut first, and hold the line: night replay, step-week vigilance, Brain versus Bedside side by side, Follow-up card, real crypto (keep the panel with an honest label), QR pairing (pre-pair instead), Basal Check’s “possibly too high” direction. **Never cut:** the confirm step, the DEMO badges, the “no patient data” banner, the noise budget.

If the whole build is running late, the minimum credible demo is one Standing Card and one Step Watch card on the same patient, with a real doctor action and a real patient confirm.

The Family View

Recommended CUT for this hackathon. Dexcom Follow and Nightscout followers already do it, so originality is zero. It contributes nothing to the challenge. Justin’s hours are spoken for. And it contradicts the privacy pitch, because a public URL serving live glucose with a read-only token in page source, shown 30 seconds after the encryption demo, invites an awkward question. Keep `frontend/family/` as a skeleton so the repo matches the spec, and put one roadmap line on the slide. The Pi-failure fallback is not the family view; it is the backend on a laptop with `IRIN_HW=mock` and replay. Decision 2026-09-25 (docs/OPEN_ITEMS.md 27): the page is kept for the Meta entry’s Family Story Level 2, but the objection is answered rather than argued with: it now reads one last-night rollup from Irin Cloud with a revocable family bearer, so no Nightscout token sits in page source, and it is never shown in the Impiricus slot.

Named Checks

A task is done when its check passes, not when the code is written.

- Each reason-code rule, including the coverage threshold.
- Each Standing Card rule, including the rate limit, both Basal Check directions, and “not enough data yet.”
- Each Step Watch status rule and amber flag, including the boundaries -15 , 4.0% , 3 of 5 , and a recovery slope of 1.0 .

- Program interaction: starting a watch suspends Basal Check; Hypo Response merges into the watch's safety card; Standing Cards resume at graduation.
- Coverage floor produces insufficient; red still goes out under insufficient; red dedupe and the 12-hour cap; red bypasses the digest.
- No card from stale nights; demo cards always badged and never sent to a non-demo inbox.
- Silence leaves plan dates unchanged; a hold shifts all later steps; holds stack.
- A doctor message never applies without confirmation; decline and expiry apply nothing; a message from a non-paired key is rejected.
- Crypto round trip; tampered ciphertext rejected; the relay stores only ciphertext; the resources-request schema has no patient field.
- Pairing: expired token rejected, reused token rejected, incomplete without device confirmation, revoke stops cards.
- Recall: asked only for recorded nocturnal lows, capped at 2 per morning; unanswered by noon becomes "no answer" and never counts as felt; the rate divides by answered, not total.
- The inferred-unfelt slope rule fires on a slow recovery and not on a treated low or when carbs were logged.
- Every metric carries a confidence label; no blank rows in brain_only mode.
- The narrative validator rejects an invented number and falls back to the template.
- Step-week vigilance only raises the predicted-low threshold, never touches the actual-low alarm, and expires after 7 days.
- Idempotent catch-up: seeking forward twice creates no duplicate cards, and a reboot mid-window gives the same result.
- Everything runs on clock.py under replay acceleration, with no time.time().
- The existing /audit presence grep still passes, because the alarm-event recorder sits outside alarm logic.

Demo Plan

Main judging stays device-first with the existing run of show. The Impiricus pitch is doctor-first. Both go in demo/DEMO_SCRIPT.md.

Run of Show

Roughly two minutes for the Impiricus beat, one narrative arc: detect, decide, verify, then the pharma moment.

1. **Pairing.** The device shows a QR code, a judge scans it, both screens show the same 4 digits, Chris confirms. The judge is now the doctor.
2. **Detect.** Replay Chris's real, date-shifted two weeks before a basal increase. The Basal Check card arrives: rise on the clean nights, the excluded nights listed with reasons. Line: "Same CGM line, opposite prescriptions. Rounds knows which."
3. **Decide.** The judge taps adjust basal; the device asks Chris to confirm by voice; the inbox shows the confirmation.

4. **Verify.** Fast-forward real post-change data; the Follow-up card shows the rise flattening. “Rounds would have flagged this N weeks earlier.”
5. **The therapy start.** The judge’s inbox shows a Spark offer: new GLP-1 start. One tap starts a Step Watch. Seek to step 2.
6. **The amber card.** Low point down 22, two near-misses, one low not remembered, rough stomach 3 of 5. The judge holds the next step and reduces insulin; Chris confirms again.
7. **The pharma moment.** On the Hypo Response row, no glucagon on file. The judge taps access and resources, picks the category and a brand, and the mock Ascend handoff appears with the banner. Flip to “what Impiricus sees”: ciphertext only.
8. **Close.** “Every hold is a patient who didn’t quit, and every one of these cards started from the patient, not from pharma. No glucose value ever left the doctor’s hands.”

If time is short in the room, beats 2 to 4 alone tell a complete story, and so do beats 5 to 7. Rehearse both halves as standalone.

Demo Data and the Seek Problem

- **Standing Cards** run on Chris’s real history: 2 to 3 real basal increases exported as date-shifted windows, 14 days either side, avoiding the Libre gaps (mid-February to late-March 2025, two weeks mid-June 2025). Reason codes on history are INFERRED and labeled so. Never present inferred history as logged.
- **Step Watch** runs on a SYNTHETIC titration scenario, labeled everywhere: a generated CSV plus a companion JSON with the plan, symptom checks, injection logs, recall answers, and alarm events, with insulin sensitivity drifting faster than the reduction at step 2, rough-stomach days after each step, and two treated predicted lows. If a teammate or consenting volunteer is on a GLP-1, real date-shifted data replaces it and the pitch gets materially stronger.
- The side-by-side Brain versus Bedside comparison needs logged context, so it uses the synthetic scenario or nights logged live during the hackathon.

Engineering note. A 24-week scenario cannot be played: $24 \times 7 \times 24 = 4,032$ hours, and $4,032 \div 60 = 67$ hours of wall time at 60x. The replay engine gains a seek that bulk-loads readings to a timestamp, advances clock.py, and lets the idempotent catch-up generate every card that should exist by then. Demo panel buttons read “jump to step N day D.” Without this the demo stalls, the same way Scenario 9 stalls without clock.py.

Q&A Ammunition

- **“Doesn’t this flood doctors?”** The noise budget: one card per type per 14 days, one per step, green in a digest, and review time may be billable. It replaces unstructured portal messages rather than adding to them.
- **“How is this different from Clarity or Tidepool?”** They show what the numbers did. Rounds explains why, removes the confounded nights, surfaces near-misses nothing else can see, measures alarm response and awareness, and delivers a decision instead of a dashboard.
- **“GLP-1s don’t cause many lows.”** Correct, and the consensus says why: clinicians reduce insulin proactively. This is how that reassessment happens between visits, and it watches the opposite risk too.

- **“Isn’t self-report weak evidence?”** It is the current clinical standard, and the two standard questionnaires disagree about the same patient nearly 60% of the time. We ask about one specific low, the next morning, with the curve attached, and the device confirms it objectively where it matters.
- **“Who buys the device?”** Nobody has to. Rounds runs on any CGM feed. Irin Bedside is the confirmatory step doctors recommend, and during a titration it can be a clinic loaner.
- **“Can Impiricus see patient data?”** No, and the panel shows it. They would still sign a BAA, because encryption is about trust, not exemption.
- **“What if someone spoofs a dose change?”** Replies are authenticated with the paired doctor’s key, the relay cannot forge, and the patient must still echo-confirm.
- **“What if the AI invents a number?”** Code computes every number and a validator rejects any that is not in the metrics.
- **“Is this a medical device?”** No card contains a dose recommendation. Production would go through FDA clinical decision support review and BAAs.
- **“Isn’t one patient too little evidence?”** Yes. n of 1, observational, retrospective, presented as proof of concept.
- **“Predicting lows isn’t new.”** Agreed. The value is room-scale waking, context, two tiers, and Rounds.
- **“Why would device makers pay when they have portals?”** Portals are pull. Rounds pushes decisions into a channel doctors already use, including the digitally dark ones portals never reach.
- **Numbers to memorize.** 8.7% glucagon fills; 17.5% IAH with about 4x severe lows; questionnaires disagree ~60%; 25 to 35% insulin reduction on GLP-1s; reassess within 2 to 3 days and at every 4-week step; over half stop within a year; \$396 per patient per year from 95251; about 6.3M type 2 insulin users, roughly 3x the type 1 population; 13 touchpoints per tirzepatide watch.

Compliance Language and Honest Limits

- Resources by category first, doctor chooses the brand, always pulled, never pushed, labeled as manufacturer resources, off for off-label use.
- Never say “sell more insulin” or “sell more GLP-1s.” Say under-titration, access, glucagon underprescribing, persistence, tolerance, safe titration.
- Persistence lift is a hypothesis Impiricus would measure, not a result.
- Historical reason codes are inferred. Everything is one patient, observational. Say “since the change,” never “the change caused.”
- Glucagon status is patient-reported; Irin cannot see prescriptions. Tolerance, injection, and recall data are patient-reported too.
- Rounds never prescribes; e-prescribing stays in the EHR.
- No pharma funding of patient devices, which raises anti-kickback problems. Clinic-owned loaners or device-maker programs are the safe versions.
- Irin remains a companion to CGM alarms, not a replacement and not a medical device.

Risks

1. **Scope.** Two programs plus crypto plus an inbox, on top of the core device, is the largest risk in this document. Mitigations: the build order puts shared foundations first, the cut order is written in advance, and the freeze rule keeps the core demo safe.
2. **Demo legibility.** Two programs can confuse judges. Mitigation: one narrative arc, and each half rehearsed standalone.
3. **Synthetic titration data.** Mitigation: Standing Cards run on real data, and the three measured numbers come from real history.
4. **Check-in and recall compliance.** Mitigation: the waiting-questions chip, one morning notification, and reporting non-answers as non-answers.
5. **Thresholds tuned on one person's data.**
6. **A judge who knows the titration apps or the clinic dashboards.**

Open Items

1. Can one project enter both main judging and the Impiricus challenge? Resolved 2026-09-25: yes, both entries are allowed; one system, three doors (Irin, Rounds, Night Buddy + Family Story).
2. Is any teammate or consenting volunteer on a GLP-1?
3. Chris lists therapy-change dates with confidence notes in gitignored ml/data/, choosing 2 to 3 away from the Libre gaps.
4. Decide: a real relay and crypto, or in-process with honest labeling. Resolved 2026-09-25: the real relay on Vultr (Atlas, real crypto), with the in-process fallback.
5. Decide whether to write Rounds into the main documents: a spec section, the contracts additions, each teammate's plan, the new invariants, the /review and /audit checklists, and the demo script. Resolved 2026-09-25: written into every main document (docs/OPEN_ITEMS.md decisions 26 and 27).
6. Compliance nuance if asked: a type 1 patient with obesity on an obesity-indicated product is on-label for weight, but glucose-management messaging is still off-label, so the rail stays off by default.
7. Reconcile this document's patient narrative with the user-perspective story written in another session, so only one version exists. Resolved 2026-09-25: the main spec's narrative is the one version, with this document's Basal Check → confirm → Follow-up arc and Maria's full watch.

Beyond the Hackathon

- **Weekly basal insulin watches** as Awikli launches, the same engine on a different therapy start.
- **Rounds for type 2 insulin users** as the main commercial push.
- **Automated insulin delivery users**, where the card becomes about settings rather than doses.
- **Other connected devices** publishing the same card format: smart pens, pumps, blood pressure cuffs, inhalers.
- **Adverse-event drafting** as a future card type.
- **A read-only family view**, gated the same way as the doctor's view.

- **Real evidence.** Everything here is one patient and retrospective. The next step is a small prospective cohort at one clinic, measuring time to insulin adjustment, severe lows, glucagon prescribing, and persistence at 6 months against matched patients.

Hosting and Sponsor Integrations (added 2026-09-25)

Everything Irin owns in the cloud runs on one Vultr VPS from `deploy/docker-compose.yml` behind Caddy with automatic HTTPS on the team's domain, `irin-out-of-sleep-at-hackgt.tech`: the relay at `api.irin-out-of-sleep-at-hackgt.tech` (this document's courier plus the hub, the Night Buddy directory and the owner-pairing routes; MongoDB Atlas; ciphertext and profile fields only, never a glucose value) and Irin Cloud at `cloud.irin-out-of-sleep-at-hackgt.tech` (the patient's own service: readings and events forwarded outbound from the Pi into Tiger Cloud hypertables, continuous aggregates that draw the app's My Irin dashboards, the family rollup, ElevenLabs audio). Nightscout stays on Railway with its database on Atlas. The Pi keeps SQLite and works with no cloud at all.

The two services are deliberately separate so the panel can prove the courier holds only ciphertext: dashboards never decide a card (every card number still comes from `nights.py`, and a test asserts the SQL rollups agree with it where they overlap), and nothing in the relay's database is a reading.

The patient app is one hosted app at `irin-out-of-sleep-at-hackgt.tech` with four tabs (Irin Device, My Irin, Irin Buddy, Irin Rounds), gated by the PIN with no login; the Irin Rounds tab holds Share with my doctor, the pairing status, the card preview, the glucagon toggle, the recall cards, the check-ins, and the step timeline exactly as the Patient App requirements say. Pairing the device to the app is the kiosk's QR and 6-digit code (single use, 10 minutes, shown only while the radar sees someone): the Pi registers it with the relay (`POST /v0/device/pairings`), the app redeems it (`POST /v0/device/pair` with the code and a username) and receives `device_url` and a token, and the Pi confirms on its next relay poll; it is the separate `DevicePairing` model, while `Pairing.peer_kind` stays `doctor | buddy`.

Narratives go through `narrative.py`'s chain of four links, Backboard → Meta Model API direct → the direct Claude call → deterministic template, with the validator after every link and Backboard's per-scope memory (patient, doctor-patient pairing, buddy pair; Readonly for family). `NARRATIVE_ROUTING` picks the provider per task, eight tasks: `clinician_card` and `morning_report` go to a Sonnet-class Claude; `buddy_line` and `family_story` to Muse Spark (through Backboard's OpenRouter provider if it carries it, else meta / muse-spark-1.3); `match_explanation`, `emergency_script` and `availability_parse` to muse-spark-1.3 direct with structured output; `second_opinion` to openai, a different provider from the writer. The device speaks its confirm screens in an ElevenLabs voice after the tone, never instead of it; the morning report gets a play button.

Python 3.14 everywhere, the Pi through uv, code kept 3.13-compatible; Cursor runs beside Claude Code under the same constitution. Full detail: the main spec (Web App, Deployment and Hosting, Tech Stack), `docs/plans/chris.md` (I-steps and C-steps), and `docs/OPEN_ITEMS.md` decisions 27 and 29.

Appendix A: Pydantic Models

```
class NightRecord(BaseModel):
    night_date: date
    window_start: datetime
    window_end: datetime
    coverage_pct: float
    reason_codes: list[str]
    code_source: Literal["logged", "inferred", "unknown"]
    rise_mgdl: float | None
    dawn_rise_mgdl: float | None
    low_point_mgdl: float | None
    tbr_pct: float | None
    minutes_below_70: float
    near_miss_count: int
    level2_count: int
    alarm_event_ids: list[str]
    is_demo: bool

class AlarmEvent(BaseModel):
    event_id: str
    tier: Literal["predicted_low", "actual_low", "stale", "high"]
    started_at: datetime
    acknowledged_at: datetime | None
    ack_source: Literal["device", "app"] | None
    escalated: bool
    rearm_count: int
    crossed_actual: bool
    presence_during: Literal["home", "away", "unknown"]

class LowEvent(BaseModel):
    low_event_id: str
    night_date: date
    started_at: datetime
    nadir_mgdl: float
    nadir_at: datetime
    minutes_below_70: float
    auc_below_70: float # mg/dL-minutes
    recovery_slope: float # mg/dL per min, 30 min after nadir
    carbs_logged_within_30min: bool
    inferred_unfelt: bool
    alarm_event_id: str | None

class LowEventRecall(BaseModel):
    low_event_id: str
    asked_at: datetime
    answered_at: datetime | None
    answer: Literal["felt_and_treated", "woke_no_symptoms",
                    "dont_remember", "was_awake"] | None
    # unanswered after the morning window = "no answer", never "fine"
```

```

class SymptomCheck(BaseModel):
    date: date
    gi: Literal["fine", "rough", "cant_eat"]
    is_demo: bool

class TitrationStep(BaseModel):
    index: int                # 0 = starting dose
    dose_label: str           # "2.5 mg", display only
    planned_start: date

class WatchOptions(BaseModel):
    ketone_prompts: bool = False
    step_week_vigilance: bool = False
    vigilance_offset_mgdl: float = 10.0    # predicted-low 70 -> 80

class TitrationPlan(BaseModel):
    plan_id: str
    drug_class: Literal["glp1", "gip_glp1", "weekly_basal"]
    drug_label: str           # display only
    steps: list[TitrationStep]
    started_at: date
    on_insulin: bool
    options: WatchOptions
    status: Literal["pending_confirm", "active", "graduated", "ended"]
    is_demo: bool

class SignalCard(BaseModel):
    card_id: str
    schema_version: str = "0"
    program: Literal["standing", "step_watch"]
    kind: Literal["basal_check", "hypo_response", "follow_up",
                  "early_check", "step_check", "step_gate",
                  "safety", "graduation", "baseline_note"]
    status: Literal["green", "amber", "red", "insufficient"]
    flags: list[Literal["lows", "highs", "awareness", "tolerance",
                        "level2",
                        "rearm", "ketone_risk", "baseline_thin",
                        "dose_mismatch", "missed_injection",
                        "rise_high", "rise_low"]]
    confidence: dict          # metric -> measured | reported |
inferred
    source: Literal["irin_bedside", "irin_brain"]
    patient_pseudonym: str
    plan_id: str | None
    step_index: int | None
    period_start: date
    period_end: date
    headline: str
    metrics: dict

```

```

    nights: list[dict]                                # {date, reason_code, code_source,
rise}
    excluded_counts: dict
    tolerance_days: list[dict]
    narrative: str
    allowed_actions: list[str]
    resource_categories: list[str]
    is_demo: bool
    generated_at: datetime

class Pairing(BaseModel):
    device_id: str
    doctor_id: str
    doctor_display_name: str
    doctor_pk: str
    status: Literal["pending", "paired", "revoked"]
    confirmed_at: datetime | None

class DoctorMessage(BaseModel):
    message_id: str
    plan_id: str | None
    kind: Literal["plan_create", "plan_update", "proceed",
"hold_step",
                    "insulin_change", "note", "schedule_request",
                    "end_watch", "dismiss"]
    hold_weeks: int | None
    insulin: Literal["basal", "bolus"] | None
    new_units: float | None
    start_date: date | None
    plan: TitrationPlan | None
    text: str | None
    created_at: datetime
    status: Literal["pending", "confirmed", "declined", "expired"]

```

Appendix B: Relay API

- POST /v0/spark/new_rx (demo only, key-gated): simulates the Spark trigger, creating an offer in a doctor's inbox. Carries no patient identity.
- POST /v0/pair and GET /v0/pair/{token}.
- POST /v0/cards with recipient_id, sender_id, nonce, ciphertext, source, and an X-Source-Key header.
- GET /v0/inbox/{recipient_id}?since= with a bearer token per inbox.
- POST /v0/messages and GET /v0/device/{device_id}/messages.
- POST /v0/resources/request with doctor_id, category, brand. The only thing a pharma-side system ever sees, and the schema has no patient field.
- GET /v0/log: the “what Impiricus sees” view.

Pairing data travels in the URL fragment, which browsers never send to servers, so neither Caddy nor the relay ever logs a token or a public key. Tokens are 128-bit, single use, and expire

in 10 minutes on clock.py. After pairing, both directions use authenticated encryption (crypto_box with the sender's secret key and the recipient's public key), so the relay can neither read nor forge. Added 2026-09-25: the base URL is <https://api.irin-out-of-sleep-at-hackgt.tech>; the device key travels as X-Source-Key; the same service also serves the Night Buddy directory routes (POST /v0/users, GET /v0/users/search, POST /v0/match) and the owner-pairing routes (POST /v0/device/pairings from the Pi, POST /v0/device/pair from the app, DELETE /v0/device/pair), and holds no readings.

Appendix C: Sources

Impiricus:

- Products: <https://www.impiricus.com/our-products/>
- Homepage: <https://impiricus.com/>
- Press: <https://www.impiricus.com/press/>
- Ascend launch (Hashmi, Qureshi, Firozvi, mission statement): <https://www.impiricus.com/impiricus-launches-ascend-an-ai-platform-to-ethically-connect-physicians-and-pharma/>
- Medvantx integration, September 2026 (Hashmi): <https://www.globenewswire.com/news-release/2026/09/08/3357938/0/en/impiricus-integrates-medvantx-to-expand-real-time-patient-access-and-therapy-support.html>
- QPharma integration, August 2026 (Hashmi, Aftab): <https://www.globenewswire.com/news-release/2026/08/25/3350955/0/en/impiricus-launches-qpharma-integration-to-power-real-time-compliant-hcp-sampling.html>

GLP-1 titration and insulin:

- 2026 consensus, adjunct GLP-1 and GLP-1/GIP RA in T1D: <https://journals.sagepub.com/doi/10.1177/15209156261449879>
- EASD 2025 tirzepatide in T1D real-world data: <https://touchendocrinology.com/insight/easd2025-real-world-data-suggest-tirzepatide-reduces-insulin-needs-in-type-1-diabetes/>
- Upcoming guidance on GLP drugs in T1D: <https://www.medscape.com/viewarticle/upcoming-guidance-address-glp-drugs-type-1-diabetes-2026a10007x5>
- Over 50% stop GLP-1s within one year: <https://www.medscape.com/viewarticle/real-world-study-finds-over-50-stop-glp-1s-within-1-year-2025a1000obm>
- Truveta ISPOR 2025, drivers of discontinuation: <https://www.truveta.com/blog/research/ispur-2025-real-world-temporal-and-indication-specific-variation-in-drivers-of-glp-1-ra-discontinuation/>
- Basal insulin use with GLP-1 RAs (Diabetes Spectrum): <https://diabetesjournals.org/spectrum/article/29/3/152/32779/Basal-Insulin-Use-With-GLP-1-Receptor-Agonists>
- Deprescribing insulin when starting GLP-1 RAs: <https://www.sciencedirect.com/science/article/pii/S2666602222000805>

- ADA Standards of Care 2025:
https://diabetesjournals.org/care/article/48/Supplement_1/S181/157569/9-Pharmacologic-Approaches-to-Glycemic-Treatment
- Awiqli approval: <https://www.healio.com/news/endocrinology/20260327/fda-approves-onceweekly-basal-insulin-for-adults-with-type-2-diabetes>

Hypoglycemia, awareness, and alarms:

- Glucagon prescribing 2011 to 2021 (Diabetes Care):
<https://diabetesjournals.org/care/article/46/3/620/148260/Glucagon-Prescribing-and-Costs-Among-U-S-Adults>
- IAH prevalence, about 17.5% with roughly 4x severe lows (PLOS One):
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0297601>
- Gold versus Clarke disagreement (Hypo-METRICS post hoc):
<https://pubmed.ncbi.nlm.nih.gov/42115542/>
- Assessing IAH in the CGM era (DOM 2025): <https://dom-pubs.onlinelibrary.wiley.com/doi/10.1111/dom.16284>
- Nocturnal hypoglycemia in the CGM era (JDST 2024):
<https://journals.sagepub.com/doi/10.1177/19322968241267823>

Market, billing, trust, and competitors:

- CDC National Diabetes Statistics Report: <https://www.cdc.gov/diabetes/php/data-research/index.html>
- SingleCare insulin statistics, citing CDC and ADA:
<https://www.singlecare.com/blog/news/insulin-statistics/>
- Dexcom Stelo over-the-counter clearance: <https://investors.dexcom.com/news/news-details/2024/Stelo-by-Dexcom-First-Glucose-Biosensor-to-be-Cleared-by-FDA-as-Over-the-Counter/default.aspx>
- CGM and RPM CPT codes: <https://www.sweetspot.health/post/understanding-cpt-codes-for-remote-cgm-monitoring>
- CMS 2026 remote monitoring updates:
<https://www.nixonlawgroup.com/resources/cms-finalizes-2026-remote-monitoring-reimbursement-updates-what-changed-for-rpm-and-rtm>
- ADA Overcoming Therapeutic Inertia: <https://professional.diabetes.org/professional-development/overcoming-therapeutic-inertia>
- Rock Health data-sharing survey: <https://www.healthcaredive.com/news/consumer-health-data-sharing-survey-rock-health/710774/>
- Willingness to share health data (PMC):
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12005618/>
- Tidepool TIDE: <https://www.tidepool.org/providers/tide>
- Glooko population health dashboard: <https://glooko.com/new-glooko-diabetes-population-health-dashboard/>
- Dexcom Clarity for clinics: <https://provider.dexcom.com/products/dexcom-clarity>
- Accu-Chek SmartGuide for MDI: <https://www.theglucoseneverlies.com/accu-chek-smartguide-mdi-cgm-series/>

- Glooko titration system: <https://www.glooko.com/press-release/fda-clears-glookos-long-acting-insulin-titration-system/>
- My Dose Coach: <https://diatribe.org/diabetes-medications/fda-clears-my-dose-coach-app-optimize-basal-insulin-dosing>
- Hygieia d-Nav: <https://www.fiercebiotech.com/medtech/fda-clears-hygieia-s-gps-for-insulin-diabetes-management-app>
- Insulia: <https://dtxalliance.org/products/insulia/>

Appendix D: Design History

1. **Irin Clinic** (first idea): a triage panel ranked by risk, an AI pre-visit brief, a resources rail, QR hand-off, FHIR export. Dropped: the Impiricus part was a bolt-on sidebar and the triage panel depended on data only one patient has. Kept: the privacy rule, the QR idea, reuse of the morning-report pipeline, resource categories rather than brands.
2. **Irin Rounds** (second idea): three cards (Basal Check, Hypo Response, Follow-up) over an encrypted channel. Strong insight, but as originally scoped it was too large, its pharma moment arrived late and was mostly glucagon, it read as remote patient monitoring, its free-form dose path would worry a pharma compliance audience, and its HIPAA claim was wrong.
3. **Rejected along the way:** Irin data as a signal for Spark or ION (violates the trust findings); a coverage evidence packet; a titration or dosing assistant (medical dosing advice, overlaps Impiricus's calculators); ambient visit recording.
4. **Step Watch** (third idea): Rounds' rejected "Therapy Watch" card promoted to a program and triggered by a new prescription, which is a real Spark event, in the drug class the challenge text names first.
5. **The merge, chosen here:** Rounds is the platform, the channel, and the card format. Standing Cards and Step Watch are two programs on it. They share the ledger, the metrics, the delivery path, the inbox, and the privacy design, and the interaction rules in "How the Two Programs Interact" keep them from talking over each other.
6. **Refinements from review:** confidence labels instead of blank rows; nocturnal burden and inferred unfelt lows; the morning recall question and the unfelt-low rate, which give a Brain-only patient an awareness metric nobody ships; the display-glances / app-interacts split; the family view cut; the corrected HIPAA language; and the replay seek, without which a 24-week scenario cannot be demonstrated.